

SumoBot! Blue Mantis

Final Report

Matthew Brady

Bella Sifuentez

Jonah Stuart

2026



Project Sponsor: OSU School of MIME, Curtiss-Wright

Faculty Advisor and Instructor: Dr. Oman

DISCLAIMER

This report was prepared by students as part of a university course requirement. While considerable effort has been put into the project, it is not the work of licensed engineers and has not undergone the extensive verification that is common in the profession. The information, data, conclusions, and content of this report should not be relied on or utilized without thorough, independent testing and verification. University faculty members may have been associated with this project as advisors, sponsors, or course instructors, but as such they are not responsible for the accuracy of results or conclusions.

ABSTRACT

This capstone project involves designing and building a competitive 3 kg remote-controlled sumo robot for competition at the Engineering Expo in June 2026. The robot must contain complex geometry aligning within Robogames Unified Sumobot rules for a "pushing match" against competitors competing within a 154 cm diameter circular ring called a dohyo. Competition scoring follows Unified Sumo Robot rules where the first team to win two rounds receives two Yuhkoh points and wins the match.

Analysis of existing sumo robot designs reveals common successful strategies including frontal ramps for pushing opponents, treads for increased traction, sharp angle ramps for wedging underneath opponents, and low centers of gravity for stability. Control methods typically involve either wireless RC transmitters or autonomous sensors. Our sponsor relayed this project will be using a strictly remote control system with no sensors.

The goal for our project is to stay in the dohyo ring while forcing the opponent out. To achieve this, we engineered a robot with a custom offensive ramp-claw mechanism that will lift and destabilize competitors while maintaining strong forward traction. Our design emphasizes durability, controlled maneuvering, and strategic geometry to gain an advantage in close-quarters combat. Through iterative design, analysis, and testing, the project demonstrates the integration of mechanical design strategies, manufacturing feasibility, and control strategy required to produce a competitive sumobot.

ACKNOWLEDGEMENTS

The completion of this project would not have been possible without the support and guidance of many individuals and organizations. Our team would like to express sincere appreciation to the Oregon State University School of Mechanical, Industrial, and Manufacturing Engineering (MIME) for providing the resources, knowledge, and environment necessary to pursue this project. We extend special thanks to Dr. Sarah Oman, our project sponsor and primary point of contact, for her continued feedback, direction, and commitment to our team's success.

We also gratefully acknowledge Curtiss-Wright and their team of engineers for serving as our manufacturing sponsor. Their willingness to collaborate, provide machining expertise, design feedback, and fabricate our final components played a critical role in bringing our design to completion.

Finally, we thank everyone else who has offered assistance, shared knowledge, or provided encouragement throughout the duration of this project. Their contributions and support have been greatly appreciated throughout the project and do not go unnoticed.

TABLE OF CONTENTS

[Use Word to delete the following Table of Contents and insert a new TOC. Include front matter (except for the cover page), body of the report, and all appendices. The Table should include three levels of headings, e.g., down to 2.3.1, as illustrated below.]

Contents

DISCLAIMER	1
ABSTRACT	2
ACKNOWLEDGEMENTS	3
TABLE OF CONTENTS	4
1 BACKGROUND	2
1.1 Introduction	2
1.2 Project Scope	2
2 DESIGN PROCESS	3
3 DESIGN PROPOSAL – First Term	5
4 Design Solution	6
4.1 Description of Solution	6
4.2 Project Results	6
5 LOOKING FORWARD	7
6 CONCLUSIONS	8
7 REFERENCES	9
8 APPENDICES	10
8.1 Appendix A: Descriptive Title	10
8.2 Appendix B: Descriptive Title	10

1 BACKGROUND

1.1 Introduction

This project centers on the design and fabrication of a remote controlled sumobot, a small compact robot inspired by traditional sumo wrestling that competes in a series of organized pushing matches of three minutes with other sumobots. These robots can be autonomous or remote controlled, also known as RC. Similar to human sumo wrestling, these sumobots attempt to force each other out of a circular arena, known as a dohyo. The project at hand is to create an original, strictly RC sumobot with a non-standard geometry, meaning not a box, that fits the customer requirements and Robogames Unified Sumobot rules and objective is to “fight” by applying a pushing force upon another opponent forcing them out of the dohyo ring.

The project objectives include developing practical engineering skills through iterative design, manufacturing, and testing within a team environment. Though we have found a manufacturing sponsor, our objectives still stand as we integrate mechanical and electrical systems for reliable performance and apply design principles that are feasible to be machined as we work together with a professional engineering team gaining insights within our project.

The project is sponsored by Dr. Sarah Oman and the School of Mechanical, Industrial, and Manufacturing Engineering (MIME) at Oregon State University. The project is also supported by Curtiss-Wright, the team’s manufacturing sponsor, who will produce the final mechanical components that our team has designed. Curtiss-Wright's involvement connects our academic design work with professional manufacturing practices exposing our team to industry standards and ensuring high quality fabrication.

This project is of strong interest to both sponsors because it bridges academic design and professional manufacturing. For Dr. Sarah Oman and the School of MIME, it reinforces experiential learning and innovation in mechanical design, giving our team of students hands-on exposure to robotics and fabrication. For Curtiss-Wright, it provides an opportunity to collaborate with emerging engineers and demonstrate advanced machining capabilities through the production of precision components. The project benefits all stakeholders by strengthening the connection between education and industry as we gain technical proficiency and teamwork experience, while the sponsors and university showcase the engineering and creativity that this project displays at Expo. Beyond these outcomes, the project addresses contemporary issues in robotics and automation by exploring efficient mechanical design and control strategies that mirror modern industrial automation challenges. It also promotes sustainable and efficient manufacturing practices through material and machining selection and design optimization.

1.2 Project Scope

The scope of this project is to design and fabricate an original RC sumobot that will compete in a series of “head to head” pushing matches against our competitor sumobots while staying within the Unified Sumobot rules. The problem the team is addressing is the need for a non-standard and high performing sumobot design that will demonstrate originality and mechanical efficiency while meeting customer requirements and strict competition rules. Most existing sumobots use simple box geometries; our team aims to innovate by developing a unique form in between the rules grey areas that improves pushing power, traction, and control stability while still being within the guidelines.

Out of scope items for this project include any autonomous operation, liquids, weapons, intentional damaging mechanism, and materials that exceed weight or machinability limits. The project as stated numerous times is strictly RC, aligning with customer expectations and competition rules.

This scope represents the team’s unified goal to design a competitive, manufacturable, and innovative

sumobot that demonstrates engineering skills, collaboration, and real world application of design and manufacturing principles while also providing some wins during the matches at Expo.

2 DESIGN PROCESS

Our team followed a traditional product development process consisting of research, design, prototyping, and manufacturing cycles. This approach allowed us to brainstorm many design alternatives while retaining the ability to iterate based on testing results. The research phase involved analyzing previous sumo robot designs to identify successful strategies including frontal ramps, tread-based traction, sharp-angle wedging geometry, low centers of gravity for stability, and RC control reliability versus autonomous sensors.

To satisfy the requirement for at least one defensive or offensive mechanism beyond a basic ramp, the team brainstormed several supplemental features. Concepts included an aluminum bar attached to a servo for swiping opponents to the side, various lifting mechanisms inspired from battle bots, and alternative ramp geometries for improved wedging capability. The team evaluated these alternatives based on their effectiveness at neutralizing opponent resistance, manufacturability within the project timeline, and compliance with competition rules regarding push based competition without flipping.

The design proposal was based around being as innovative as we could without breaking any rules. After discussion with Dr. Oman, it was noted that there was a large grey area in the unified sumo rules that could be exploited to give an advantage. The rules are very specific about flipping, but there is no stated documentation regarding lifting the opposing bot. After discovering this, the team did a lot of research on how we could design a mechanism that could efficiently lift the opposing bot and carry it out of the ring.

The final design concept centers on a dual mechanism lifting system consisting of a bottom ramp assembly and a top clamping arm. The bottom mechanism utilizes two high torque servos mounted in parallel at the front of the chassis, controlling an angled ramp blade designed to slide underneath the opponent bot during initial contact. The innovation in this design is the integration of extruded metal support bars extending from each servo arm allowing to provide force at the front of the ramp lip. We also located a set of front wheels that have two purposes, they allow for the bot to turn and drive more effectively, and also act as a block to provide a reaction force driven into the ground to support the ramp lifting an opposing bot.

The top mechanism consists of a servo actuated claw arm positioned above the ramp assembly. This arm engages after the ramp has successfully positioned itself under the opponent, securing the lifted bot and preventing escape. This approach provides several advantages over traditional pushing strategies. By lifting the opponent off the ground, their drive wheels lose traction, reducing their ability to resist. The clamping mechanism ensures the opponent cannot disengage, a common failure mode in ramp reliant push designs.

The drive system features four DC gear motors driving large rear wheels with wide silicone tires for maximum traction, and uses a pivot style control system to change directions quickly to maneuver around



other bots. Control is implemented through an RC transmitter interfaced with an Arduino Uno, providing precise control during multiple received inputs from all subsystems. Manufacturing considerations focused on feasibility from complexity and considering weight restraints. The best combination we could optimize relies on a 3D printed chassis using standard PLA, a ramp that was created using pressed and machined aluminum, a claw that was pressed from aluminum, and lastly a servo moment are 3D printed in a SLA resin bath.

3 DESIGN PROPOSAL – First Term

This chapter details the prototyping design process used to develop the robot, including key design considerations, iterative modeling, and physical testing. It highlights how performance priorities such as traction, stability, and simplicity informed design decisions, while also introducing the evolution of both passive and active mechanisms through successive prototypes.

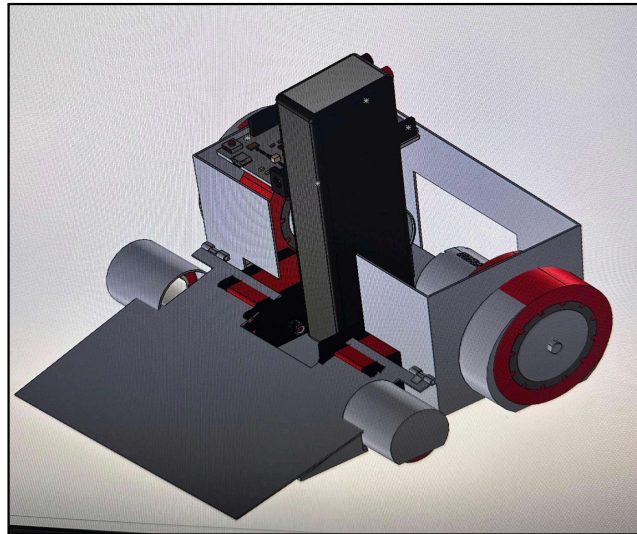


Figure 1: Early CAD model prototype of our components inside a chassis of specified size. (Created in Solid Works)

Robotics competitions have existed for a long time and thus there is a lot of information online about these robots. Successful sumo robots prioritize traction, low center of mass, and prioritize torque over speed or maneuverability. Robots that focus on complex mechanisms over simple advantages can suffer from more issues such as reliability, where more simply designed bots are less prone to failure.

Some previous approaches use four wheel drive systems with high torque motors to maximize their pushing force. High friction tire materials are used to maximize traction against the metal competition ring surface. Minimizing wheel slip is very important. If a robot slips it leads to reduced pushing force or even being pushed out itself.

Chassis geometry also plays a very important role. Low profile designs reduce the likelihood of being lifted or destabilized by an opponent by minimizing the distance a force can act and cause a moment reaction on the robot un-settling it. Many teams use a wedge shaped front end to attempt to gain an advantage. Competition rules restrict ramps, limiting this strategy's maximum potential. Teams must identify an effective chassis geometry which will defend and apply sufficient pushing force on an opponent.

Teams are required to have at least one moving offensive or defensive mechanism in this competition, so teams must navigate the possibilities and pair a strong or complimentary active system with their passive

design. A wedge, varying side geometry, asymmetric design, and low surface area would all be examples of passive systems. An active system would use a motor or spring mechanism to move something on the bot to push the opponent, like a flap or arm. Passive mechanisms are easier to integrate than active ones. Active mechanisms introduce complexity and introduce failure modes, but could provide more of a tactical advantage if designed well.

In the context of this project, many existing solutions are not directly transferable due to different vehicle classes or design philosophies. Many parts of existing robots can be re-formed into a new and compliant design. Additionally, the limited development timeline limits the experimentation and re-design process. While some professional robots may rely on advanced materials and expensive electronic components, the competition during the Engineering Expo will be among teams with the same budget and schedule constraints. So we expect the other bots in competition to be of similar parts we have access to.

Our design process is focused around rapid prototyping and revisions. We meet each week to work on aspects of our prototype for the following week. We try to have a new chassis 3D modeled and 3D printed chassis for every Monday meeting so we can place our components inside to make sure they fit and so we can plan revisions. This approach to prototyping helps us quickly make new design decisions and modifications.

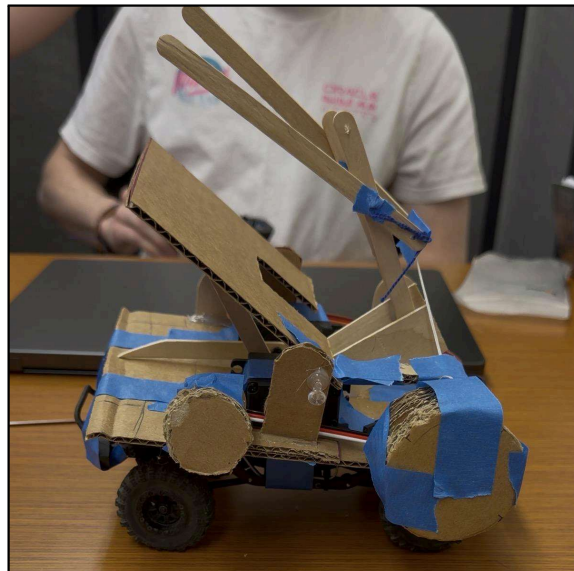


Figure 2: Early prototype showing ramp and claw concept mechanism

The striking feature component of this design will be an intimidating ramp and claw mechanism (see Figure 2, lower ramp made of cardboard and upper claw made of popsicle sticks). On the front of the design will be a ramp which can be actuated up and down. Above this ramp will be two claw like features that can also be actuated up or down. The ramp is intended to slip just underneath an opposing robot and lift it, either slightly to reduce their traction so we can push them off, or fully into the air so we can set them outside the ring. The claw mechanism on top is the key part of this design. The claw will come

down to pinch or hold a robot in the ramp mechanism so we can overpower them. When working together, these features of the design will be a force to be reckoned with. One without the other would not be anywhere as strong, but when both are used in harmony the competition stands no chance.

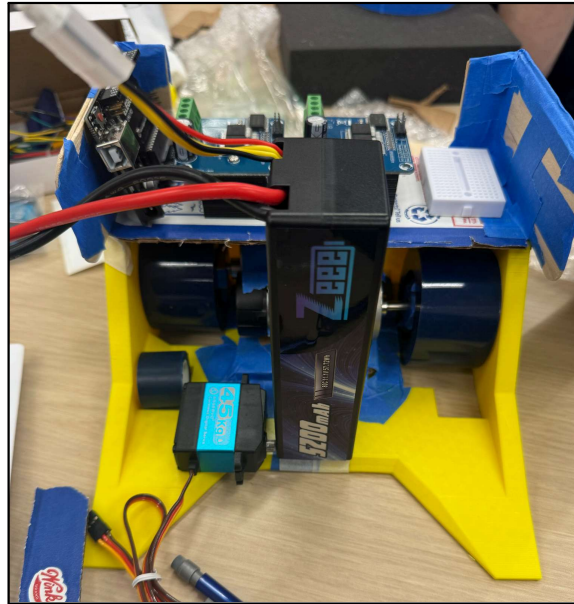


Figure 3: Proof of concept physical prototype chassis with components installed. Assortment of materials used, including the main 3D printed base support, cardboard and popsicle sticks.

The proof of concept shown in Figure 3 was our initial design showing how our components would fit inside the bottom half of our chassis. We learned a lot from this process, mainly we decided where we were going to place our components in the next iteration. The big stressor was the battery which we decided to place upright in this model. After performing another proof of concept we found that we could design a shelf like structure to hold components more efficiently and we decided to lay the battery flat on the third shelf. These shelves will help us keep the components and wires close to each other and minimize our robots surface area. The concept in Figure 3 greatly helped us iterate on our design and 3D model with a new version with the shelves and component spacing.



4 Design Solution

[Use this section to discuss the final design your team implemented for this project. Provide detail at all times regarding the functionality of the solution, how you proved the system met all Customer Requirements and Engineering Specifications, and the final results of your project test results.]

[Include a brief introduction to this section here before moving on.]

4.1 Description of Solution

[Description here.]

4.2 Project Results

[Detailed discussion of your project results from testing and client interactions.]

5 LOOKING FORWARD

[This section can also be used to provide detailed information to your client regarding how they can finish the project, improve upon it, etc. Include observations you may have on the overall project such as ways to improve the capstone experience for future teams or future clients.]

Use this section to describe any and all information that future teams for this project would need to know to continue this project, create a second iteration of this project, or otherwise. Reference how the results of the testing completed may influence design changes in the future. This section should be detailed and thorough.]



6 CONCLUSIONS

Provide a general conclusion here of the project, team goals, the results, etc. Then complete the following subsections.

[Provide a commentary on how your team applied engineering design principles to produce a solution that meets specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors. Your reflection does not need to address every category stated in the previous sentence but should provide details regarding those most relevant to your project. What were those factors that were most important to your project (public health? Cultural factors? Etc.)? How does your design address those factors? How did you ensure your design was safe? How did you ensure your design met certain environmental issues? etc.]

7 REFERENCES

[Include here all references cited, following the reference style described in the Style Guide. There should only be one Reference list in this report, so all individual section or subsection reference lists must be compiled here with the main report references. If you wish to include a bibliography, which lists not only references cited but other relevant literature, include it as an Appendix.]



8 APPENDICES

[Use Appendices to include lengthy technical details or other content that would otherwise break up the text of the main body of the report. These can contain engineering calculations, engineering drawings, bills of materials, current system analyses, and surveys or questionnaires. Letter the Appendices and provide descriptive titles. For example: Appendix A-House of Quality, Appendix B- Budget Analysis, etc. **All Appendices should start on a new page.**]

8.1 Appendix A: Descriptive Title

8.2 Appendix B: Descriptive Title